databricks

Continuous Integration (CI)

LECTURE

Planning the Project

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In this lecture, you'll learn how to plan a data visualization project by structuring environments, managing data across dev, stage, and prod, and setting up a CI/CD pipeline in Databricks for smooth, secure deployments.

Planning the Project

Requirements

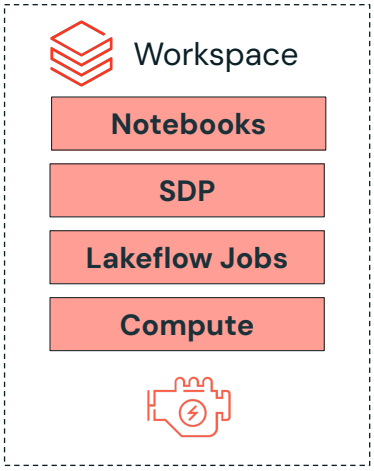
Deliverable

- Visualize Health Data

Tasks

- Ingest daily incremental CSV files to a bronze table
- Create a clean silver table
- Create gold tables to share with consumers

Databricks Assets



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This project focuses on visualizing health data by following a structured data pipeline. We begin by ingesting daily incremental CSV files into a Bronze table, then refining this raw data into a clean Silver table. From there, we create a summarized Gold table, which is shared with consumers and used to build the final visualization. Throughout the process, we make use of various database assets, including notebooks, Spark Declarative , Lakeflow Jobs, and compute resources, to ensure a smooth and efficient execution.

Planning the Project

Setting Up Your Data Environments

Dev Data

- Often a Small Static Subset of Production Data
- Can be Anonymized or Synthetic Datasets
- Supports Rapid Development and Testing
- Ensures Privacy and Data Integrity

Stage Data

- Staging Data Mirrors Production Structure & Volume, Typically Static
- Can be Anonymized or Scrubbed Sensitive Information
- Ensures Realistic Testing and Validation

Prod Data

- Production Data: Live & Fully Operational
- Contains Real User Data
- Continuously Updated
- Requires High Security, Privacy, & Compliance Standards

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Within a CI/CD pipeline

In development, data is often anonymized or generated using synthetic datasets to allow rapid development and testing without compromising privacy or production data integrity.

If you have an staging environment, staging data should closely mirror production in structure and volume, with anonymized or scrubbed sensitive information to ensure realistic testing and validation.

Production data is live, fully operational, and continuously updated, containing real user data, and must be handled with high security, privacy, and compliance standards.

Each environment should have data suited for its specific purpose, balancing realism, security, and compliance at every stage. How this is implemented will depend on your organization and the sensitivity of your data.

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Isolating Environments

Workspaces

Utilizing multiple workspaces, one for each environment

DEV

STAGE

PROD

Catalogs

Utilizing multiple catalogs, one for each environment

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Isolating your environments for the different stages of development is key to developing a CI/CD pipeline. This ensures that code is developed and tested in the developing and staging prior to touching the production environment.

The minimal setup is to have two environments “Development & Stage” and “Production”, but this can vary depending on your organizations requirements.

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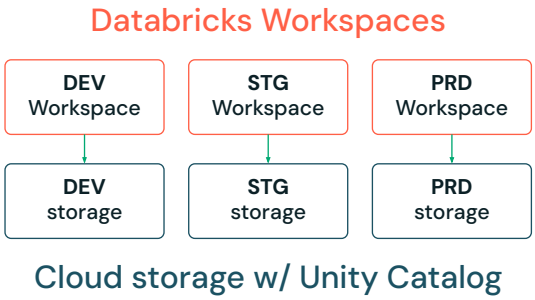
Workspace Isolation Overview

You can **isolate** your dev, stage and prod environments at the Workspace and storage level.



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Within Databricks, one methods for isolating your environments includes creating a separate Workspace for the dev, stage and prod environments. This ensures that all development is isolated from your production environment.

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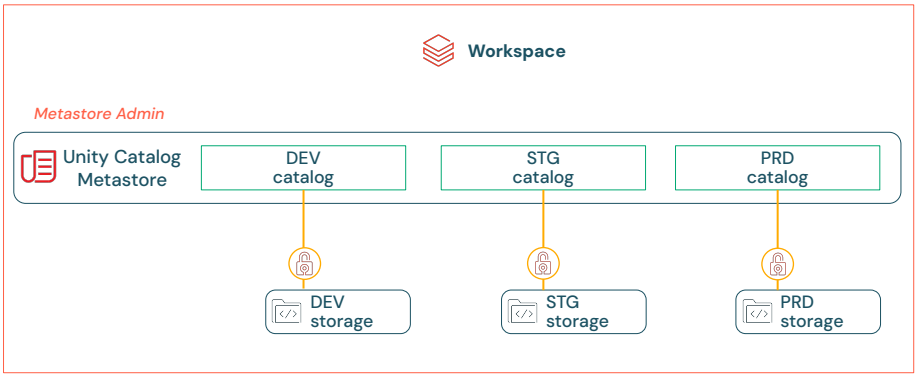
Unity Catalog Isolation

Storage isolation

This example separates the storage locations on catalog level

UC Access Control

Users should only gain access to data/ metadata based on agreed access rules



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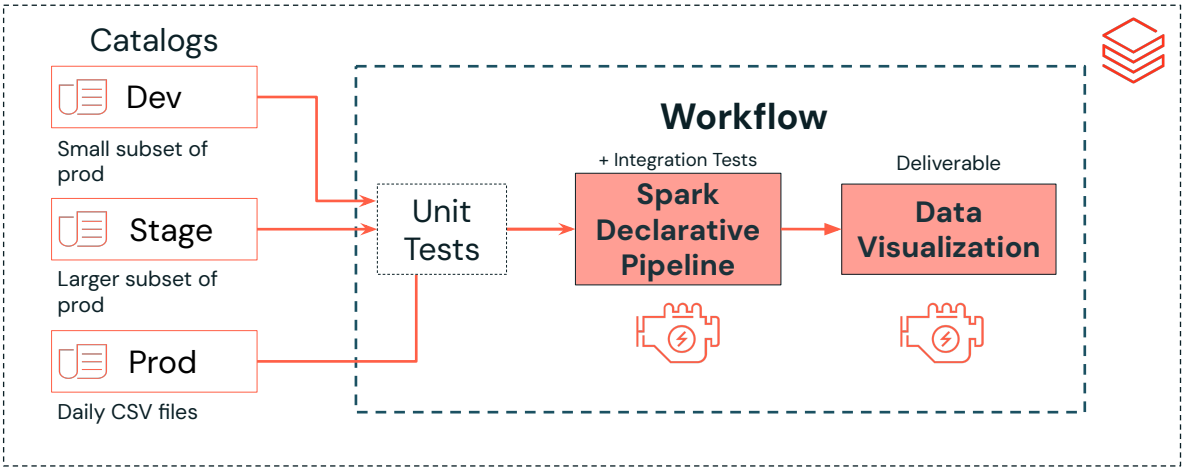
Another method for isolating your environments within Databricks is Unity Catalog isolation.

With this method, you create a catalog for each(dev, stage and prod). The dev environment includes the dev data, stage the staging data, and prod the production data.

With this method you can also utilize Unity Catalog access control for your developers, only giving them the required permissions for each.

Planning the Project

Course Project Architecture



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Now that we have an understanding of our project, our data, and how to isolate environments, let's dive into the project setup.

In this project we will create a catalog for each environment: dev, stage, and prod.

The dev catalog will contain a small, static subset of our production data, used for development and initial testing.

The stage catalog will hold a larger subset of production data, enabling more comprehensive testing as we move through our CI/CD process.

Finally, the prod catalog will contain the live production data that we rely on for final operations.

Our workflow starts by developing on the dev data, running unit and integration tests as we progress through the pipeline. The pipeline is set up with Databricks Lakeflow Jobs, which execute the unit tests, Spark Declarative pipeline, and final visualizations.

As we test the pipeline through each stage, we will ensure everything is functioning correctly before deploying to production.

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Thank you for completing this lesson and continuing your journey to develop your skills with us.